# DECISION MAKING & DISTRACTION

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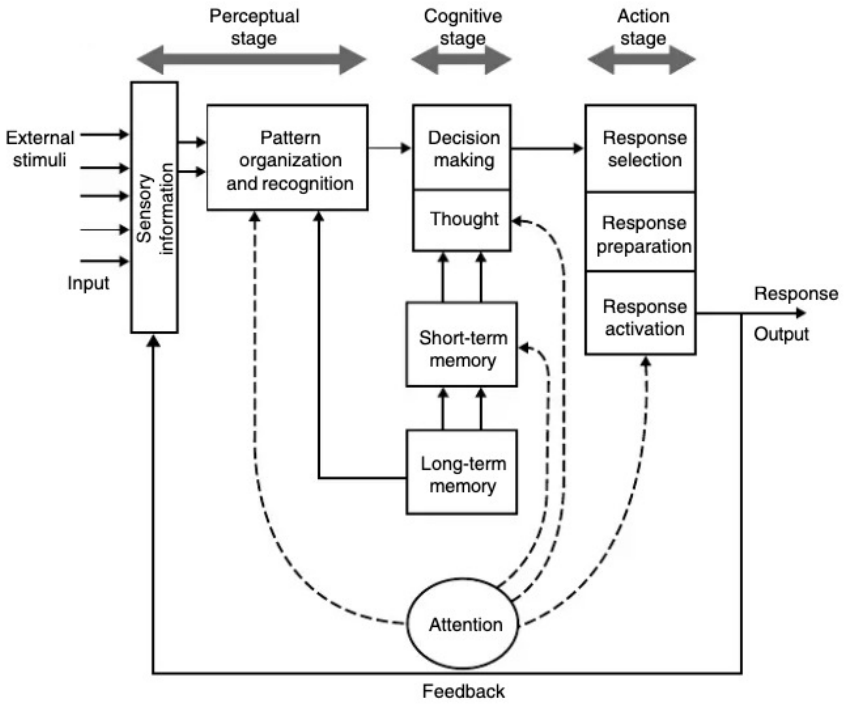
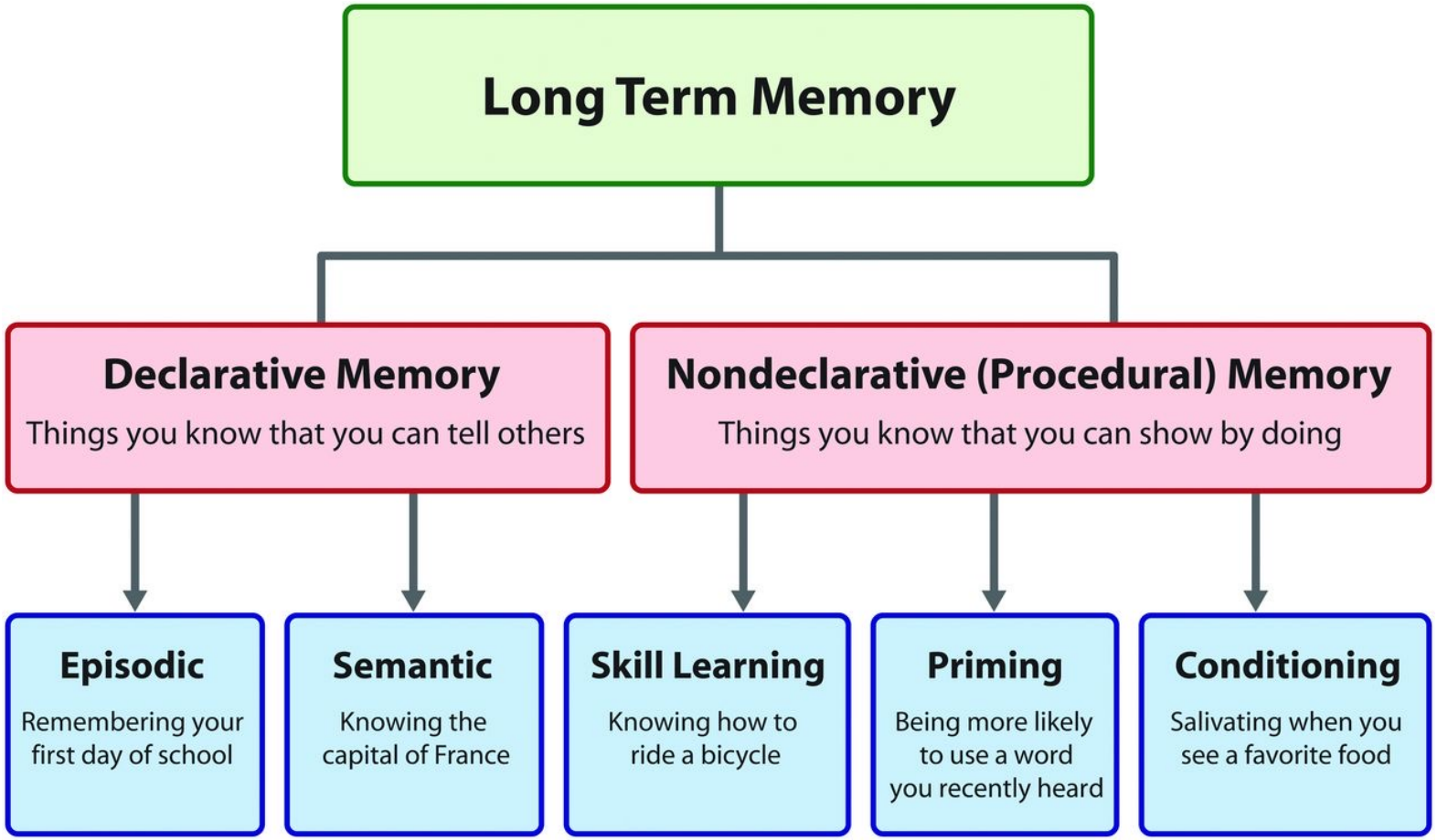
ENP-0112

# HUMAN INFORMATION PROCESSING MODEL

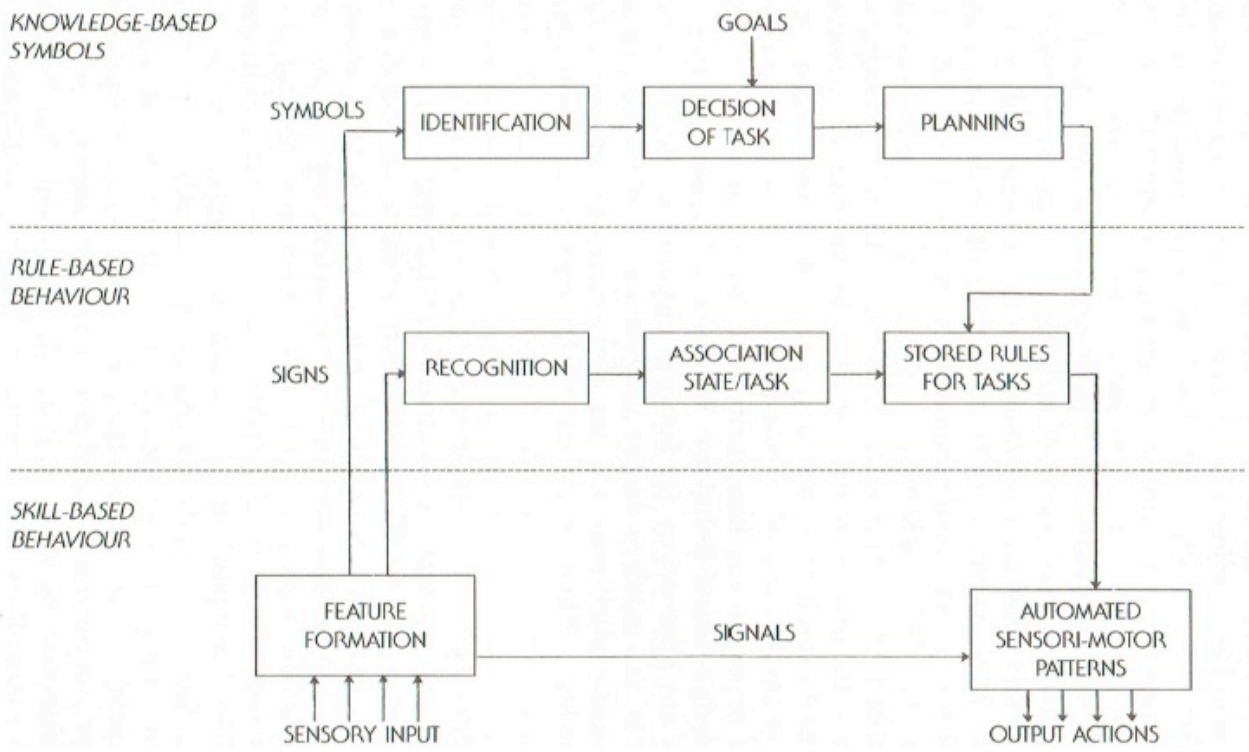



Proctor & Van Zandt, 2018




F.D. Raslau (2014)

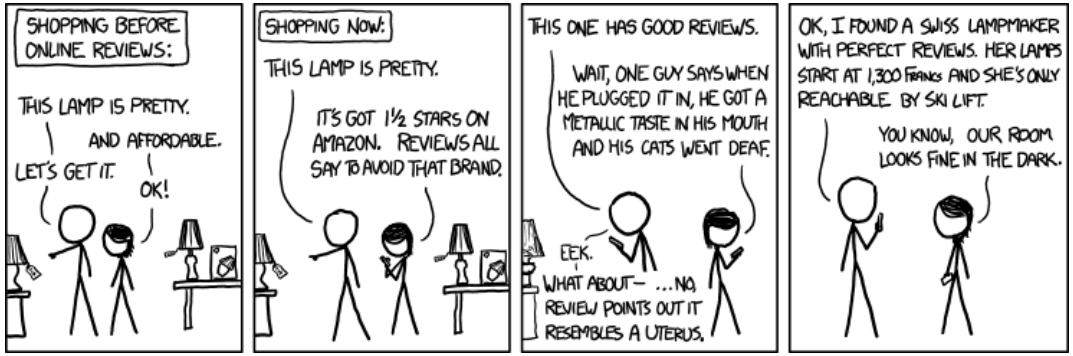
# RASMUSSEN LEVELS OF INFORMATION PROCESSING




Rasmussen, 1983

# DECISION MAKING FRAMEWORK

- Normative Decisions
  - How decisions *should* be made - Utility
  - Prescriptive, specifies choices to make under ideal circumstances
  - Limited working memory constrain these models
- Problems:
  - Humans are not rational decision makers - Bounded rationality (Simon, 1957)
  - No universal agreement on the worth associated with various outcomes




xkcd.com/1036

# DECISION MAKING FRAMEWORK

**Naturalistic Decisions**

- How decisions are *actually* made
- Try to explain how people actually think
- How do we deviate from normative rationality?

**Characteristic settings for naturalistic decisions**

- Uncertain, ill-defined dynamic environments
- Shifting, ill-defined, or competing goals
- Action/feedback loops
- Time stress
- High stakes
- Multiple players
- Organizational goals and norms

# NATURALISTIC DECISION MODELS

- Use heuristics – problem solving methods that are not necessarily optimal or rational, but are sufficient for reaching an immediate goal (“Satisficing”)
- All options may not be considered
- Account for resource limitations
  - Incomplete information
  - Uncertainty in information
  - Time constraints
  - Memory limited (retention and recall)
  - Attention limited
- May be biased

# KAHNEMAN'S TWO SYSTEMS MODEL OF THINKING

- System 1
  - Operates automatically and quickly
  - No sense of voluntary control
  - Quick associations
- System 2
  - Allocates resources to effortful mental activities that demand it
  - Associated with concentration and choice, self-control

Kahneman, 2011

# CONFLICT BETWEEN SYSTEMS 1 & 2

LEFT

left

right

RIGHT

RIGHT

left

LEFT

right

Kahneman, 2011

# EXAMPLE SYSTEM 1 CHARACTERISTICS

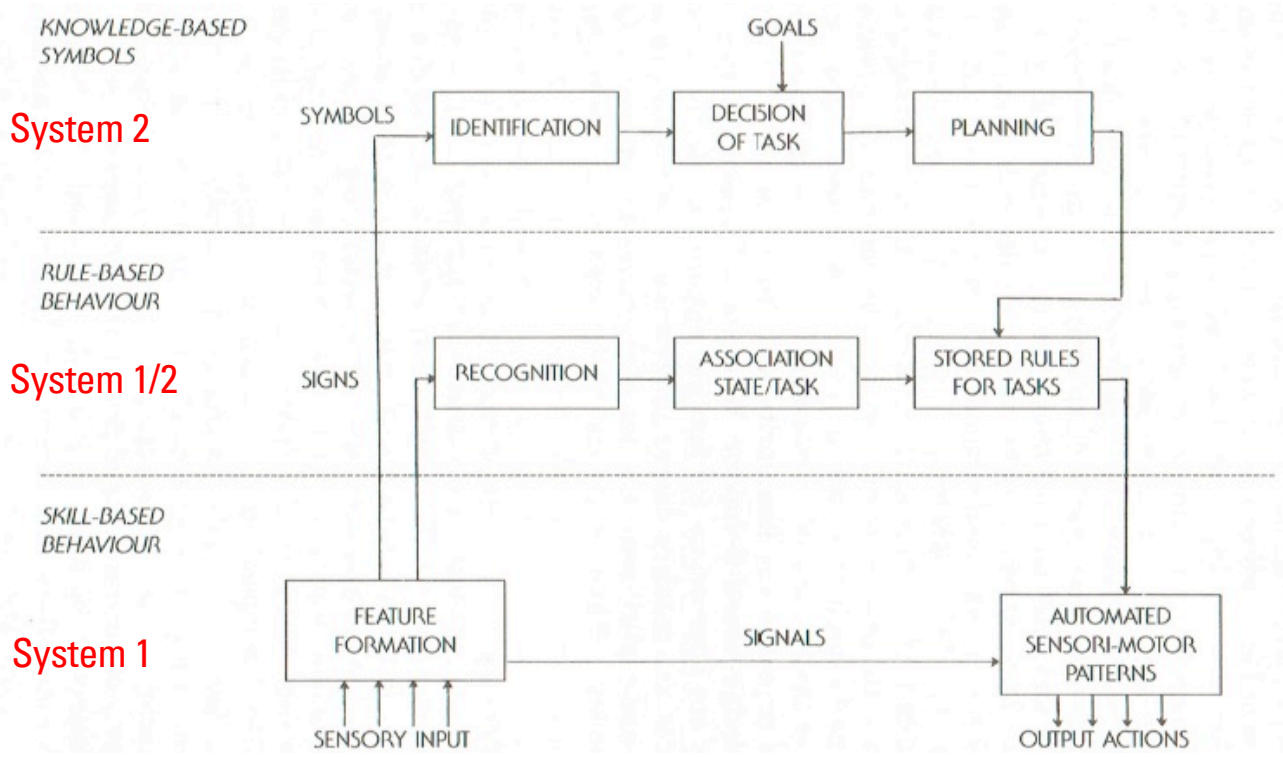
- Generates impressions, feelings, and inclinations; when endorsed by System 2 these become beliefs, attitudes, and intentions

- Executes skilled responses and generates skilled intuitions after adequate training (procedural memories)

- Is biased to believe and confirm

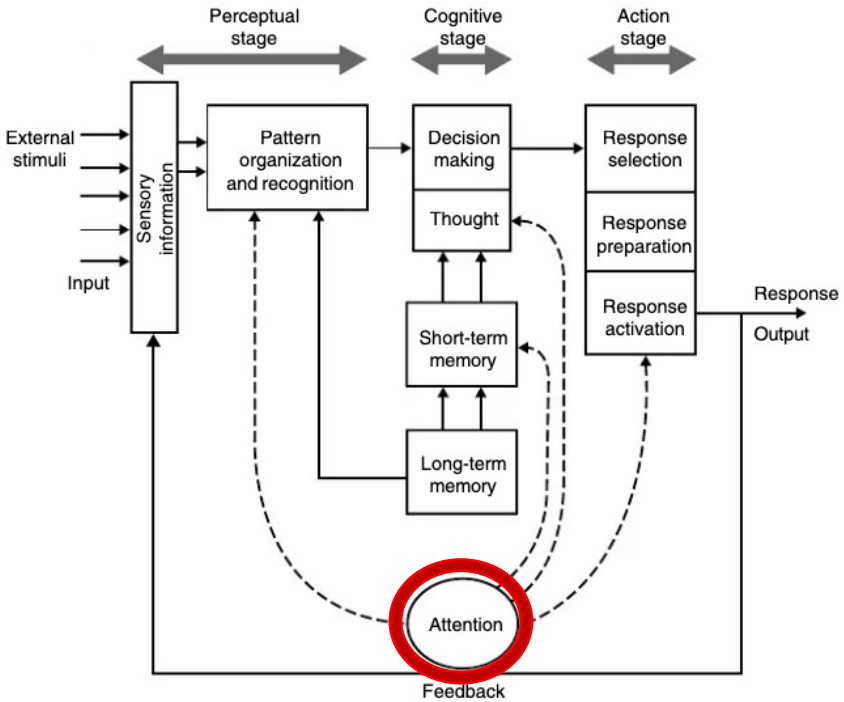
Kahneman, 2011

# RASMUSSEN LEVELS OF HUMAN INFORMATION PROCESSING




Rasmussen, 1983

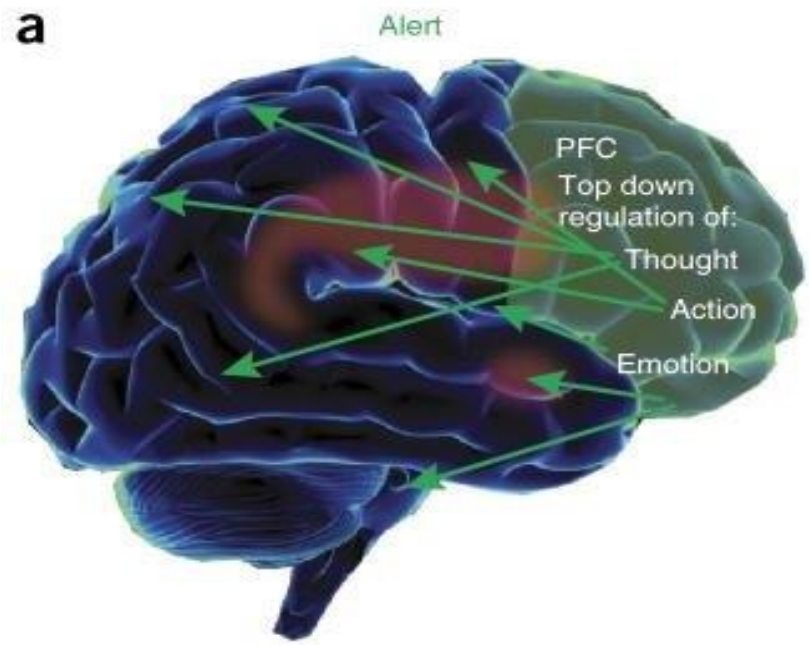
# HUMAN INFORMATION PROCESSING MODEL




Proctor & Van Zandt, 2018

# EXECUTIVE FUNCTION (PREFRONTAL CORTEX)

- Last to evolve
- Sometimes called “working memory, mental sketch pad”
- Understood recently with neuroimaging techniques
- Becomes fully mature by the early twenties
- WW 2 studies of pilots-skilled during peace time, crashed in battle due to mental errors. Stress…..
  - Impaired performance requiring complex, flexible thinking.
  - Improved performance of simple or well-rehearsed tasks (D. Broadbent,1971)




Arnsten, 2015

# ATTENTION

- ***Attention:*** Increased awareness directed at a particular event or action to select it for increased processing.
  - ***Selective Attention:*** ability to focus on certain sources of information and ignore others
  - ***Divided Attention:*** ability to attend to multiple sources of information

# NOVICE VERSUS EXPERIENCED DRIVERS

**EYE MOVEMENTS**

- Novices concentrate their fixations in a smaller area
- Novices looked closer in front of the vehicle and more to the right of the vehicle's direction of travel
- Novices sampled their mirrors less frequently
- Novices made pursuit eye movements on the freeway route while the experienced drivers made only eye fixations

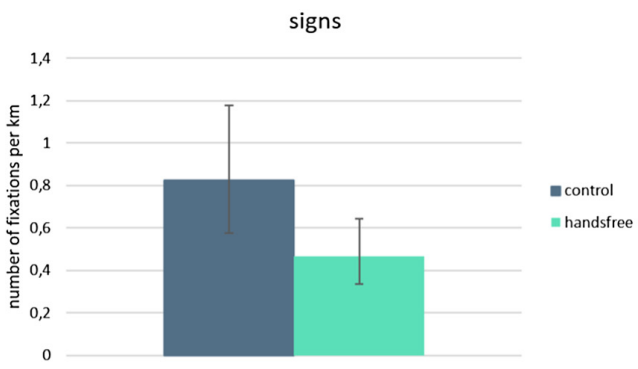
Mourant & Rockwell, 1972

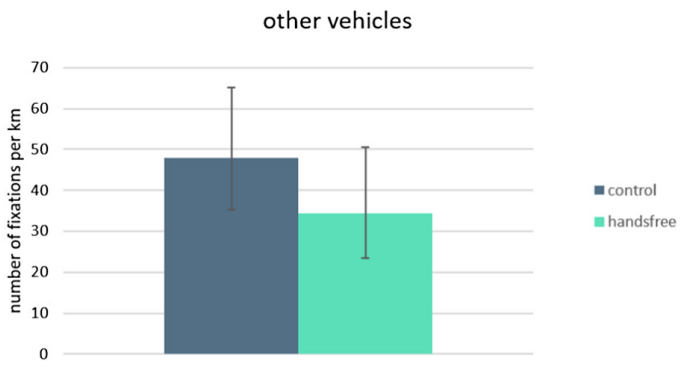
**VEHICLE CONTROL**

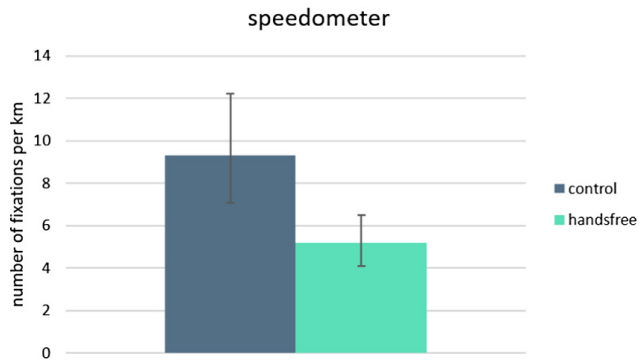
- Novices tend to show more variable performance for tasks such as lane keeping
- Novice seem to primarily control vehicle heading angle while experienced drivers also manage lane deviations.
- Control actions of experienced drivers have lower gain and show greater stability.

Smiley, Reid & Fraser, 1980

# CHANGES IN DRIVER EYE MOVEMENTS

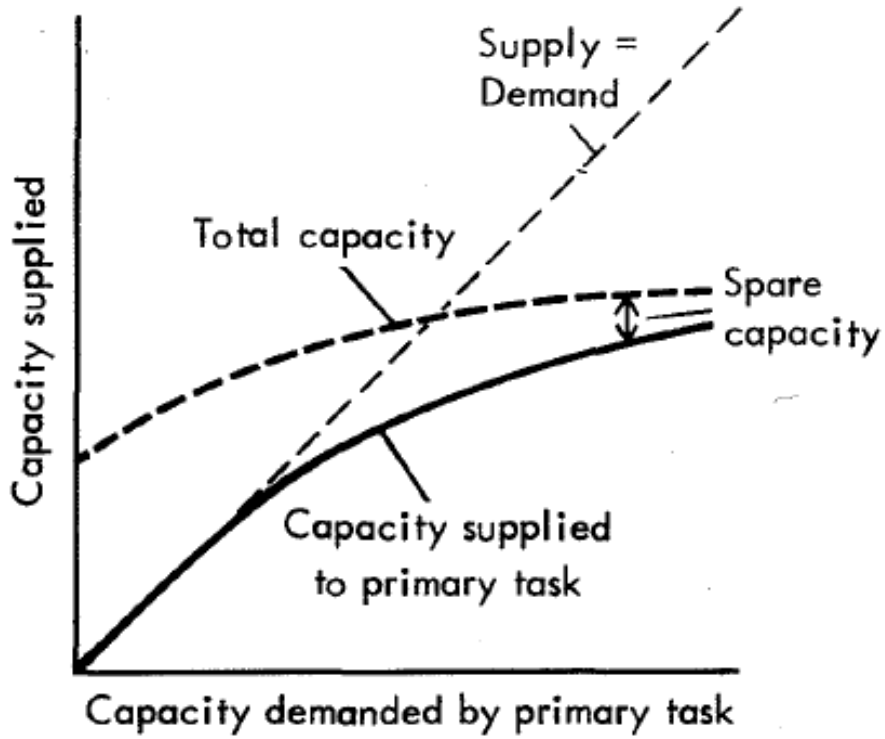



Desmet & Diependaele, 2019

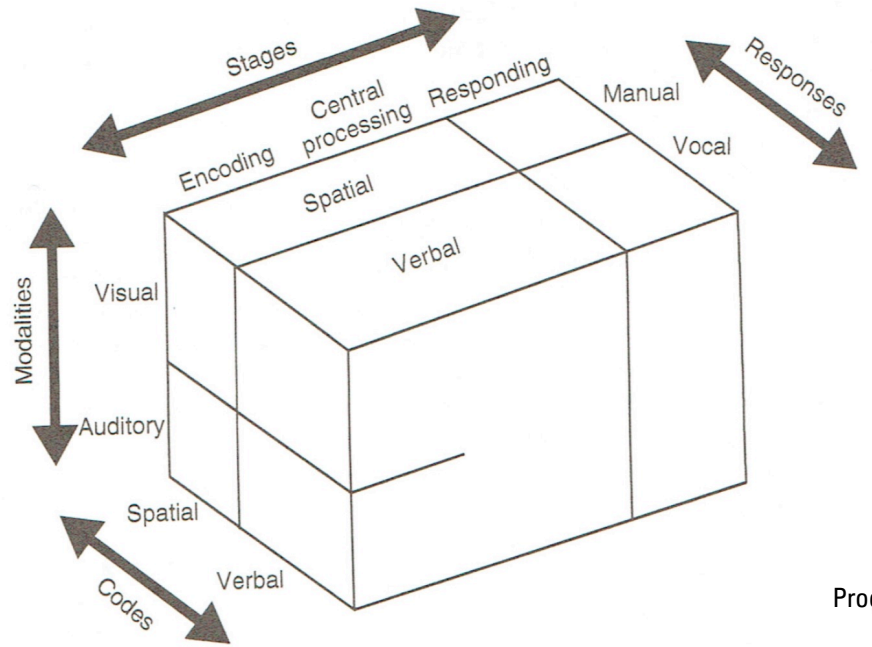
# RESOURCE MODELS – SINGLE RESOURCE MODEL




Kahneman, 1973

# RESOURCE MODELS – MULTIPLE RESOURCE MODEL

Different attentional resources exist for different sensory-motor modalities and coding domains




Proctor and Van Zandt, 2018

# ACCIDENT DRIVER TASKS

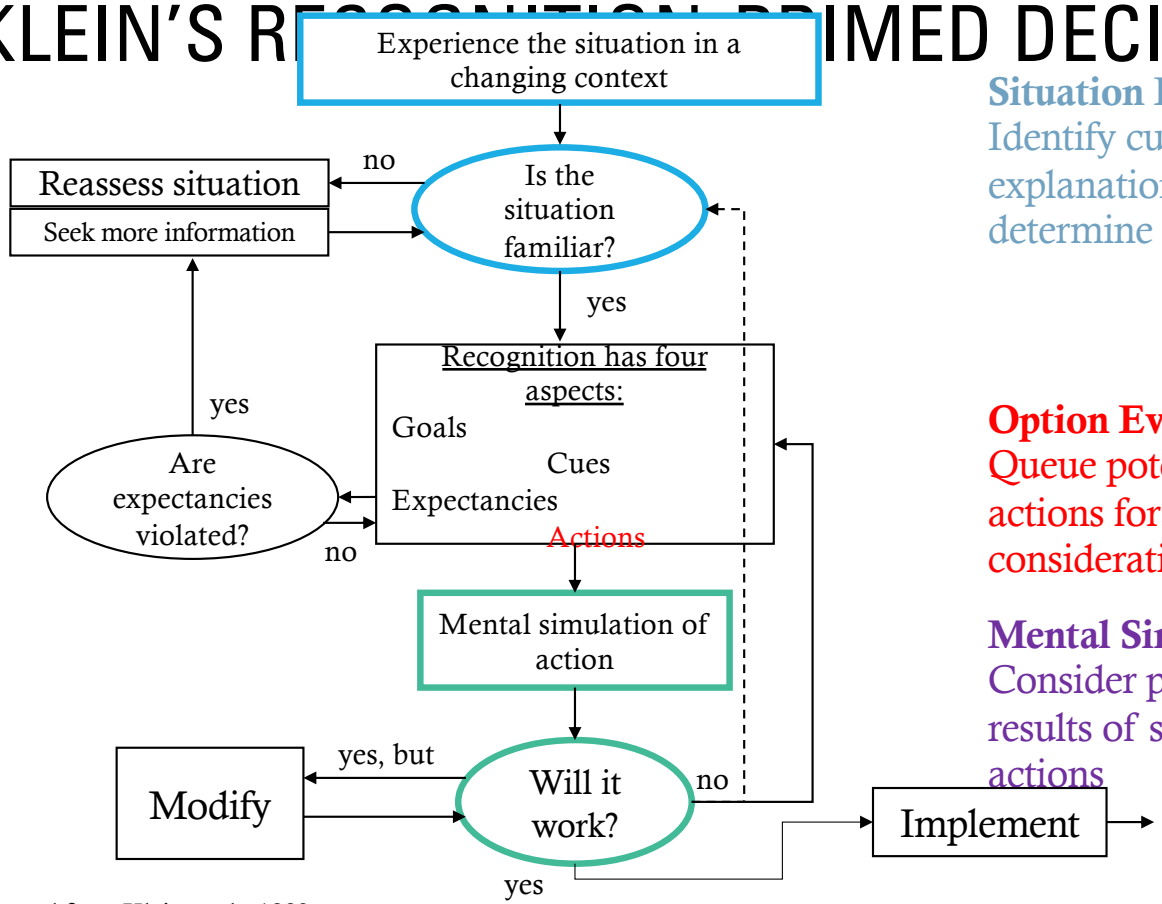
- Stay on road
- Respond to other vehicles on the road
- Talk with friend
- Keep in visual contact with lead car

# REFERENCES


- Arnsten, AFT. Stress Weakens Prefrontal Network: Molecular Insults to Higher Cognition. *Nat. Neuroscience*, 2015 October 18 (10): 1376-1385
- Desmet, C., & Diependaele, K. (2019). An eye-tracking study on the road examining the effects of handsfree phoning on visual attention. *Transportation Research Part F: Traffic Psychology and Behaviour*, 60, 549-559. https://doi.org/https://doi.org/10.1016/j.trf.2018.11.013
- Kahneman D. *Thinking, Fast and Slow*. Farrar, Straus and Giroux, New York, 2011.
- Klein G, Orasanu J, Calderwood R (Eds). *Decision Making in Action: Models and Methods*. Ablex, 1993.
- Mourant, R. R., & Rockwell, T. H. (1972). Strategies of Visual Search by Novice and Experienced Drivers. *Human Factors*, 14(4), 325-335. https://doi.org/https://doi.org/10.1177/001872087201400405
- Proctor R, Van Zandt T. (2018) Human Factors in Simple and Complex Systems 3rd Edition. CRC Press.
- Rasmussen J. Skills, rules, and knowledge; signals, signs, and symbols, and other distinctions in human performance models. *IEEE Transactions on Systems, Man, and Cybernetics* SMC13(3): 257-266, 1983.
- Simon, H. *Models of Man*, New York: John Wiley, 1957
- Smiley, A., Reid, L., & Fraser, M. (1980). Changes in Driver Steering Control With Learning. *Human Factors*, 22(4), 401-415. https://doi.org/10.1177/001872088002200402
- Underwood, G., Chapman, P., Berger, Z., & Crundall, D. (2003). Driving experience, attentional focusing, and the recall of recently inspected events. *Transportation Research Part F*, 6, 289-304.

# EXTRAS

# KLEIN'S RECOGNITION PRIMED DECISION MAKING




**Situation Recognition:**
Identify cues, consider explanations, set goals, determine possible actions

**Option Evaluation:**
Queue potential actions for consideration

**Mental Simulation:**
Consider possible results of selected actions

Iterative

Adopted from Klein et al., 1992